

Ian Ku

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PROFESSIONAL SUMMARY

Founding engineer with 3 years of experience building production full-stack, cloud-native, and AI-driven systems. I've led end-to-end development across RAG pipelines, agentic workflows, and real-time backend architectures. I move fast, take ownership, and ship reliable software that users trust.

SKILLS & CERTIFICATIONS

- ❖ **Certifications:** AWS Certified Cloud Practitioner
- ❖ **Programming Languages:** Python, TypeScript/JavaScript, C/C++, C#, Java, R.
- ❖ **Frameworks:** React, Next.js, Vue, Node.js, FastAPI, Flask, Electron, ASP.NET
- ❖ **Cloud & DevOps:** AWS (Lambda, API Gateway, CDK, S3, DynamoDB, CloudFront, IoT), Docker, GitHub Actions, Temporal, Ansible, Linux, WebSockets, OAuth, Git.
- ❖ **Database:** PostgreSQL, MySQL, DynamoDB, Firebase, Redis.
- ❖ **Artificial Intelligence:** LangChain, Pinecone, RAG, OpenAI Embeddings.
- ❖ **Speaking Languages:** Mandarin (fluent), English (fluent), Malay (advanced), Cantonese (advanced)

WORK EXPERIENCE

Stealth AI Startup **San Francisco, California**
Founding Engineer *August 2025 -- Present*

- Building a production AI application using **Node.js**, **TypeScript**, **Firebase**, and **LangChain** with **hybrid search strategies**.
- Architecting scalable backends, data models, and async **WebSocket** workflows to support multi-user AI interactions.
- Improving **RAG** quality, latency, and retrieval accuracy while developing integrations and multimodal capabilities that let AI operate across tools, devices, and user workflows

Inflow (Personal AI Startup) **Remote, USA**
Founder & Founding Engineer *Jun 2025 -- August 2025*

- Built a **LangChain-based RAG** system using **Pinecone hybrid search** to classify emails, detect phishing patterns, extract intents, and generate context-aware replies.
- Architected backend with **FastAPI**, **PostgreSQL**, **Docker**, and **Temporal** to support async, multi-user workflows at scale.
- Built the frontend in **Next.js/TypeScript** with **WebSocket**-driven inbox intelligence, dashboards, and role-based access control.

Ulendo Technologies **Ann Arbor, Michigan**
Software Engineer *April 2024 -- Jun 2025*

- Built an IoT management application using **TypeScript**, **React.js**, and **AWS Amplify**, plus a **Python/Electron** desktop app.
- Developed low-level ESP32 firmware in C++ (**ESP-IDF**, **FreeRTOS**) for real-time **UART/I2C** data collection.
- Engineered the full AWS backend: IoT pipeline with **AWS IoT Core/SQS/S3/Lambda**, high-availability APIs with **Python + AWS CDK (Lambda, API Gateway)**, and scalable **NoSQL** data models using **DynamoDB**, **GraphQL**, and **S3**.

UCI MIND (UC Irvine Institute for Memory Impairments and Neurological Disorders) **Irvine, California**
Software Programmer *November 2022 -- March 2024*

- Developed **Dockerized** external web applications with **Vue.js**, **JavaScript**, **Git**, and **Python Flask** backend.
- Independently built a tissue repository web application using **R** in **R Shiny** and **Git** with our Alzheimer's **RedCap** database.

Ansys, Inc **Remote, USA**
Software Developer - Meshing Intern *June 2023 -- September 2023*

- Independently implemented a 3D mesh solution for a PCB model provided by Apple using **Python** and **C#**.
- Developed GUI features in Ansys Mechanical utilizing **C++**, while adeptly managing version control via **Git** and **Azure Ops**.

Kheradvar Research Group (KLAB) **Irvine, California**
Full Stack Developer Researcher *July 2022 -- March 2023*

- Developed web applications in **TypeScript** and **Vue.js**, supporting cardiac MRI segmentation and analysis workflows.
- Engineered 2D/3D MRI image segmentation and visualization features using **OpenCV**, **PyVista**, **AWS S3**, and **Python**.
- Leveraged and built APIs to provide end-to-end functionalities using **FastAPI** with compute parallelization using **Celery**.

N2N Connect Berhad **Kuala Lumpur, Malaysia**
Software Engineering Intern *August 2020 -- October 2020*

- Participated in developing portfolio management web applications using **C#**, **ASP.NET Core**, and **Entity Framework**.
- Built a new in-production market data database using **PostgreSQL** and **bash** scripting, deployed using **Docker**.

EDUCATION

University of California, Irvine (UCI) **Irvine, California**
B.S in Computer Engineering | GPA: 3.61 *September 2021 -- December 2023*

De Anza College **Cupertino, California**
A.S in Computer Science | GPA: 3.59 *January 2019 -- June 2021*